

MATHEUS DE OLIVEIRA SALDANHA

Computer Scientist | Software Engineer

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in matheus-de-oliveira-saldanha

EDUCATION

- 2018-2022

B.Sc. in Computer Science

Federal University of Santa Catarina (UFSC), Florianópolis, *Grade 8.77*

Emphasis on Digital Certification and Post-Quantum Cryptography.
- 2024-2026

M.Sc. in Computer Science

Federal University of Santa Catarina (UFSC), Florianópolis, *Grade 9.00*

Emphasis on Isogeny-based Cryptography and IoT.
- 2026-2029

P.h.D. in Computer Science

Campinas State University (Unicamp), Campinas

Emphasis on Homomorphic Encryption.

WORK EXPERIENCE

- June 2025

Researcher, LABSEC, Internship

> Brazilian Civil Registry : focused on solutions to improve information management in the Brazilian Civil Registry, using advanced security and AI techniques to ensure data protection, efficiency, and regulatory compliance.
- September 2024
September 2022

Software Developer , MOTOROLA SOLUTIONS, Full time

> Emergency Call Handling Simulation : led the design and development of a minimum viable product (MVP) from the ground up, tailored to U.S. clients. The solution significantly improved the testability and reliability of emergency call handling environments.

> U.S. Jail Management System : focused on the maintenance and enhancement of inmate management software utilized by multiple counties across the country.

JavaC#AngularMongoDBAzure DevOpsDocker

September 2022
January 2022

Software Developer , CELK SISTEMAS, Full time

> Health Management Project : aimed at enhancing the quality of Brazil's public healthcare system. Involved in the implementation of large-scale systems across multiple public health agencies nationwide. Contributed as part of a newly formed team that achieved, within the first quarter, the same revenue generated over the entire previous year.

JavaGoWicket FrameworkPostgreSQL

January 2022
January 2021

Software Developer, LABSEC (UFSC COMPUTER SECURITY LABORATORY), Internship

> Digital Document Validator : focused on the healthcare sector. Developed a system for validating digitally signed documents, including electronic medical prescriptions. Redesigned the system architecture, resulting in a 50% reduction in maintenance time for affected processes.

JavaBashPostgreSQLGitLab DevOpsDocker

March 2020
November 2018

Software Developer, LABSEC (UFSC COMPUTER SECURITY LABORATORY), Internship

> Hawa Certificate Management System : focused on developing software solutions for Certification Authorities. Expanded support for digital signature methods by integrating the system with a hardware cryptographic module, enabling a broader range of secure signature algorithms.

JavaBouncyCastlePostgreSQLGitLab DevOpsDocker

LANGUAGES

Portuguese

Native

English

Fluent

PROGRAMMING SKILLS

Core

Java, Go, C#, C, C++

Scripting

Python, Sage, Bash, Octave/Matlab

Others

Lua, Haskell, Prolog, Scheme, Perl

Assembly

MIPS, ARM, x86

Web

JavaScript, Angular, CSS, HTML

Databases

SQL, MongoDB

AWARDS

- 2018-2020 **National Institute of Information Technology (ITI) Scholarship**
Awarded for contributions to the Hawa Certificate Management System developed at LabSEC.
- 2021-2022 **National Institute of Information Technology (ITI) Scholarship**
Awarded for contributions to the Digital Signature Validator developed at LabSEC.
- 2024-2026 **CAPES Master's Scholarship**
Master's studies funded by the Coordination for the Improvement of Higher Education Personnel (CAPES).
- 2024-2026 **Foundation for Support of Research and University Extension (FAPEU) Scholarship**
Awarded for contributions to the Brazilian Civil Registry developed at LabSEC.
- 2026-2026 **CAPES PhD's Scholarship**
PhD's studies funded by the Coordination for the Improvement of Higher Education Personnel (CAPES).
- 2026-2030 **FAPESP PhD's Scholarship**
PhD's studies funded by the São Paulo Research Foundation (FAPESP).

TALKS

- 2025 **Roundtable : Careers in Computing**, Jantar dos Filósofos, UFSC.
Presented insights on research careers in computing to primarily first-year undergrad students.
- 2025 **LabSECDay 25th Anniversary**, LabSEC, UFSC.
Presented the past and ongoing research on LabSEC.
- 2025 **Civil Registry Academic Workshop I**, LabSEC, UFSC.
Presented the latest projects of the Civil Registry in LabSEC.
- 2025 **Civil Registry Academic Workshop II**, LabSEC, UFSC.
Presented the latest projects of the Civil Registry in LabSEC.

TEACHING

- 2024 **Cryptography of Key Exchanges : A Brief Introduction**, Jantar dos Filósofos, UFSC.
Delivered an introductory presentation on DH and ECDH to undergraduate students.
- 2024 **Cryptography of Key Exchanges : From Classic To Post-Quantum**, SECCOM, UFSC.
Taught a mini-course on DH, ECDH, SIDH, and CSIDH key exchange protocols to undergraduate students.
- 2025 **Teaching Internship in Computer Security**, UFSC.
Assisted faculty with instructional activities in the undergraduate Computer Security course.

EVENTS

- 2019 **II Workshop on Quantum Computing**, UFSC.
Talks and Hackathon on Quantum Computing.
- 2019 **SECCOM**, UFSC.
Academic Week of Computing and Information Systems : Talks, Hackathon, and Mini-Courses.
- 2019 **The Developer's Conference**, Florianópolis.
Talks, Workshops, and Networking Events on Software Development and Technology Trends.
- 2024 **IEEE CASS Tour Santa Catarina**, UFSC.
Talks, Technical Sessions, and Networking Events on Circuit Design, Signal Processing, and Emerging Technologies.
- 2024 **SECCOM**, UFSC.
Academic Week of Computing and Information Systems : Talks, Hackathon, and Mini-Courses.
- 2025 **Fq16**, USP São Carlos.
International Conference on Finite Fields and Their Applications 2025 : Talks and Networking Events.
- 2025 **XXV Simpósio Brasileiro de Cibersegurança**, Foz do Iguaçu, PR.
Presentation of the paper *Enhancing LoRaWAN Security: Addressing Static Root Keys with Post-Quantum Cryptography*.
- 2025 **Indocrypt 2025**, IIIT Bhubaneswar, Odisha, India.
Presentation of the paper *Hardened CTIDH: Dummy-Free and Deterministic CTIDH*.
- 2026 **XXVI Simpósio Brasileiro de Cibersegurança**, Armação de Búzios, RJ.
Presentation of the paper *From LUT to Layer: Neural Network Inference with CKKS Functional Bootstrapping*.

- 2025 **Matheus Saldanha**, Alexandre Giron, Ricardo Custódio, and Thaís Idalino. 2025. Enhancing LoRaWAN Security : Addressing Static Root Keys with Post-Quantum Cryptography. In Anais do XXV Simpósio Brasileiro de Cibersegurança, setembro 01, 2025, Foz do Iguaçu/PR, Brasil. SBC, Porto Alegre, Brasil, 367-383. DOI : <https://doi.org/10.5753/sbseg.2025.11371>.
- 2025 Banegas, G., Hellenbrand, A., and **Saldanha, M.** (2026). Hardened CTIDH : Dummy-Free and Deterministic CTIDH. In : Dutta, R., De Feo, L., Gangopadhyay, S. (eds) Progress in Cryptology – INDOCRYPT 2025. INDOCRYPT 2025. Lecture Notes in Computer Science, vol 16372. Springer, Cham. https://doi.org/10.1007/978-3-032-13301-4_9
- . 2026 **Matheus Saldanha**, Guilherme Horn, Mauricio Ferreira, Andreis Purim, Alex Cassinelli, and Hilder Pereira. 2026. From LUT to Layer : Neural Network Inference with CKKS Functional Bootstrapping. In Anais do XXVI Simpósio Brasileiro de Cibersegurança, setembro 01, 2026, Armação dos Búzios/RJ, Brasil. SBC, Porto Alegre, Brasil, 1415-1421. DOI : <https://doi.org/10.5753/sbseg.2026.27731>.